

AI5090: STOCHASTIC PROCESSES

MID-TERM EXAM

DATE: 05 APRIL 2026

Instructions:

- Fill in your name and roll number on at the top of the answer booklet.
- This mid-term exam is for a total of 30 MARKS, and will count towards 30% of your total grade.
- You may use any result covered in class directly without proof.
- Hints are provided for some questions.
However, it is NOT mandatory to solve the question using the approach in the hints.
If you think you have a better approach in mind than the one given in the hint, feel free to present your approach.
- Show all your working clearly.
We want to see your thought process, and possibly provide partial credit for the intermediate logical steps.
- Plagiarism will NOT be entertained at any length.
If you are caught cheating during the exam, your answer script will NOT be evaluated.

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that all random variables appearing in the questions below are defined with respect to $\mathcal{F}$.

Basic Formulae

1. We say that $X_n \xrightarrow{\text{a.s.}} X$ if

$$\mathbb{P}(A_{\lim}) = 1, \quad \text{where } A_{\lim} := \left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \right\} = 1.$$

2. We say that $X_n \xrightarrow{\text{m.s.}} X$ if

$$\lim_{n \rightarrow \infty} \mathbb{E}[(X_n - X)^2] = 0.$$

3. We say that $X_n \xrightarrow{\text{p.}} X$ if

$$\forall \varepsilon > 0, \quad \lim_{n \rightarrow \infty} \mathbb{P}\left(\left\{|X_n - X| > \varepsilon\right\}\right) = 0.$$

4. We say that $X_n \xrightarrow{\text{d.}} X$ if

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x) \quad \forall x \in C_{F_X},$$

where C_{F_X} = set of continuity points of F_X .

5. The moment generating function of a random variable X is defined as the function

$$M_X(t) = \mathbb{E}[e^{tX}], \quad t \in \mathbb{R}.$$

6. The characteristic function of a random variable X is defined as the function

$$C_X(s) = \mathbb{E}[e^{jsX}], \quad s \in \mathbb{R}.$$

7. Taylor's series for C_X :

If $\mathbb{E}[|X|^k] < +\infty$ for some $k \in \mathbb{N}$, then

$$C_X(s) = \sum_{m=0}^k \mathbb{E}[X^m] \frac{(js)^m}{m!} + o(s^k).$$

8. Strong law of large numbers:

If $\{X_n\}_{n=1}^{\infty}$ is a sequence of i.i.d. random variables with finite mean $\mu = \mathbb{E}[X_1]$, then

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} \mu.$$

9. Dominated convergence theorem (DCT):

Suppose that $\{X_n\}_{n \in \mathbb{N}}$ is a sequence of real-valued random variables such that $|X_n| \leq Y$, $\mathbb{E}[Y] < +\infty$. Then,

$$X_n \xrightarrow{\text{a.s.}} X \quad \text{implies} \quad X_n \xrightarrow{\mathcal{L}^1} X.$$

1. Let $\{X_n\}_{n \in \mathbb{N}}, \{Y_n\}_{n \in \mathbb{N}}$ be sequences of real-valued RVs. Let X, Y be extended real-valued RVs.

(a) **(2 Marks)**

Suppose that $X_n \xrightarrow{\text{a.s.}} X$ and $Y_n \xrightarrow{\text{a.s.}} Y$. Prove that $X_n + Y_n \xrightarrow{\text{a.s.}} X + Y$.

You may assume that $X + Y$ is well-defined (i.e., not of the form $\infty - \infty$) everywhere.

(b) **(1 Mark)**

Construct an example to show that if $X_n \xrightarrow{d.} X$ and $Y_n \xrightarrow{d.} Y$, then $X_n + Y_n$ does not necessarily converge in distribution to $X + Y$.

(c) **(3 Marks)**

Let $Y_1, Y_2, \dots$ be IID random variables which are uniformly distributed on $\{0, 1, \dots, 9\}$. For each $n \in \mathbb{N}$, let

$$X_n := \sum_{k=1}^n \frac{Y_k}{10^k}.$$

Show that $X_n \xrightarrow{\text{a.s.}} X$, where

$$X := \sum_{k \in \mathbb{N}} \frac{Y_k}{10^k},$$

and identify the distribution of X .

2. Students of AI@IITH compete with one another for GPU resources in the AI Data Centre according to the following model: at each discrete round $n \in \{0, 1, 2, \dots\}$, $A_n \in \{0, 2\}$ students attempt to access the server. The server, having a large buffer space, puts the users into a queue and serves one student per round (i.e., one student is cleared from the queue in every round). Assuming that Q_n is the number of students in queue at the end of round n , we have the model

$$Q_n = \max\{0, Q_{n-1} + A_n - 1\}.$$

It is known that $Q_0 = 0$, and $\{A_n\}_{n \in \mathbb{N}}$ is an IID sequence of random variables, with common PMF

$$\mathbb{P}(A_1 = 2) = p = 1 - \mathbb{P}(A_1 = 0), \quad 0 < p < \frac{1}{2}.$$

Furthermore, it is known that $A_n \perp\!\!\!\perp Q_{n-1}$ for all n .

(a) **(3 Marks)**

Assuming that $\{Q_n\}_{n \geq 0}$ is a time-homogeneous DTMC, specify its state space and write down its TPM P .

(b) **(2 Marks)**

Is P irreducible? Aperiodic?

(c) **(1 Mark)**

Compute a stationary distribution for P .

3. Let $\{X_n\}_{n \in \mathbb{N}}$ be a homogeneous DTMC on a finite state space $\mathcal{X}$ and TPM P .

(a) **(3 Marks)**

Given a positive integer $n \geq 2$, sets $A_1, \dots, A_{n-1} \subset \mathcal{X}$, and states $x, y \in \mathcal{X}$, show formally that

$$\mathbb{P}(\{X_{n+1} = y\} \mid \{X_1 \in A_1, \dots, X_{n-1} \in A_{n-1}, X_n = x\}) = P(x, y).$$

(b) Let $\mathcal{X} = \{1, 2, 3\}$ and $P = \begin{pmatrix} 1-p & p & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ for some fixed $p \in (0, 1)$. Let $f : \mathcal{X} \rightarrow \{0, 1\}$ be defined as

$$f(1) = 0, \quad f(2) = 1 = f(3).$$

For each $n \in \mathbb{N}$, let $Y_n = f(X_n)$.

i. **(2 Marks)**

For any fixed $n \geq 2$, evaluate the probabilities

$$\mathbb{P}(\{Y_{n+1} = 0\} \mid \{Y_n = 1, Y_{n-1} = 0\}), \quad \mathbb{P}(\{Y_{n+1} = 0\} \mid \{Y_n = 1, Y_{n-1} = 1\}).$$

ii. **(1 Mark)**

Is the process $\{Y_n\}_{n \in \mathbb{N}}$ Markov? Justify.

4. **(3 Marks)**

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function.

Furthermore, let f be bounded, i.e., $|f(x)| \leq L$ for all $x \in [0, 1]$, for some fixed constant $L > 0$. For each $n \in \mathbb{N}$, let

$$B_n(x) = \sum_{k=0}^n \binom{n}{k} x^k (1-x)^{n-k} f\left(\frac{k}{n}\right), \quad x \in [0, 1].$$

For any fixed $x \in [0, 1]$, determine the value of $\lim_{n \rightarrow \infty} B_n(x)$.

Hint: Express $B_n(x)$ as the expected value of some random variable and apply DCT.

5. Let X be a random variable with finite MGF everywhere, i.e., $M_X(t) < +\infty$ for all $t \in \mathbb{R}$.

(a) **(2 Marks)**

For any fixed $x \in \mathbb{R}$, show that

$$e^{tx} - 1 \leq tx e^{tx} \quad \forall t \in \mathbb{R}.$$

Hint: Setting $y = tx$, prove that $e^y(y-1) \geq -1$ for all $y \in \mathbb{R}$.

(b) **(2 Marks)**

For each $t \in \mathbb{R}$, show that $\mathbb{E}[X e^{tX}] < +\infty$.

(c) **(2 Marks)**

Using the results of parts (a) and (b) above, show formally that

$$\left. \frac{d}{dt} M_X(t) \right|_{t=0} = \mathbb{E}[X].$$

Hint: Apply DCT.

6. Let P be an irreducible, **column stochastic** TPM on a finite state space $\mathcal{X}$, with $P(x, x) < 1$ for all $x \in \mathcal{X}$.

(a) **(1 Mark)**

Identify a stationary distribution for P .

(b) **(2 Marks)**

Consider another TPM $\tilde{P}$ defined by

$$\tilde{P}(x, y) = \begin{cases} 0, & x = y, \\ \frac{P(x, y)}{1 - P(x, x)}, & x \neq y. \end{cases}$$

Identify a stationary distribution for $\tilde{P}$.

Quiz 04

1. **(2 Marks)**

Consider a random walk on the integer line $\mathbb{Z}$ starting from 0 and having IID uniform $\{\pm 1\}$ step size at every time instant. If P denotes the TPM of the random walk, identify a left eigenvector of P corresponding to the eigenvalue $\lambda = 1$.

2. **(3 Marks)**

Consider a game in which a biased coin $C \sim \text{Ber}(0.9)$ is tossed repeatedly and independently. **At the start of the game, you are given exactly 2 lives.** Each toss of the coin results in the following outcomes: if the coin lands heads, you gain one life, up to a maximum of 3 lives. In particular, if you already have 3 lives and the coin lands heads, your lives remain unchanged. Every time the coin lands tails, you lose a life. The game continues until you lose all your lives. Determine the probability that the game terminates without you ever reaching the state of 3 lives at any point during the gameplay.